

# Preserving Interactive Audiovisual Software

An Independent Time-Based Media Conservation Study of Golan Levin's  
*Audiovisual Environment Suite* (1998–2000)

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Independent portfolio case study | 12 August 2026

**Scope note.** This is an independent educational and professional-development study based on public documentation and technical examination of a retrieved historical documentation movie. It is not an institutional condition report, does not claim access to the artwork's preservation masters or source code, and does not represent the artist, a collecting institution, or IMMA.

## Executive Summary

This study examines the conservation implications of Golan Levin's *Audiovisual Environment Suite* (AVES), a software-based artwork in which user gestures generate and manipulate abstract image and synthetic sound in real time. The project combined artwork-identity research with a practical examination of an official historical Yellowtail documentation movie retrieved from the MIT Media Laboratory AVES archive. The technical workflow used MediaInfo, FFprobe, SHA-256 fixity checking, FFmpeg decode testing and QCTools, followed by manual audiovisual review.

The principal conservation finding is that preserving AVES cannot be reduced to preserving recordings of its output. Its identity depends on executable software, real-time behaviour, user interaction, audiovisual mappings and the technical environment that enables those relationships. A second major finding is documentary ambiguity: contemporary MIT documentation describes five interactive systems, an NTT ICC exhibition page describes six, and Levin's current project archive describes seven. This discrepancy should remain visible in the conservation record until the relevant iterations and exhibition configurations can be established.

The study therefore recommends an emulation-first investigation for historical software, while preserving original binaries and disk images where legally and technically possible, documenting hardware and interaction dependencies, establishing fixity, retaining reference documentation, and validating any emulated environment against significant experiential properties rather than treating successful software launch as proof of authenticity.

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## 1 Purpose, Scope and Method

The aim was to simulate the first stages of a museum time-based media (TBM) assessment while remaining explicit about the limits of available evidence. The study used public primary or institutional sources for artwork research and did not assume that a web-accessible documentation file was an artwork master. This distinction is central: documentation can preserve evidence of an audiovisual event without preserving the generative system that produced it.

The workflow comprised: (1) provenance-led source research; (2) artwork identity assessment; (3) retrieval of a historical Yellowtail QuickTime documentation movie; (4) technical metadata extraction; (5) SHA-256 fixity establishment and verification; (6) cross-tool stream analysis; (7) full decode testing; (8) QCTools signal review and perceptual spot-checking; and (9) artwork-level dependency, significance and risk analysis. This broadly reflects museum guidance that media-art stewardship requires both conceptual and technical knowledge, component documentation, artist information, installation evidence and continuing conservation records (Matters in Media Art, n.d.).

## 2 Artwork Identification and Research Problem

|                                     |                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Artist</b>                       | Golan Levin                                                                                                                                     |
| <b>Title</b>                        | <i>Audiovisual Environment Suite (AVES)</i>                                                                                                     |
| <b>Period</b>                       | 1998–2000                                                                                                                                       |
| <b>Type</b>                         | Interactive audiovisual software / generative digital artwork                                                                                   |
| <b>Core action</b>                  | User gestures generate and manipulate abstract animation and synthetic sound in real time.                                                      |
| <b>Historical platform evidence</b> | MIT documentation states that AVES software was distributed for Windows 2000 as part of the <i>Scribble/Active Score Music</i> DVD-ROM context. |

Levin’s MIT project page describes AVES as systems built around an audiovisual “substance” controlled by user gestures, with low-level synthesis linking sound and image. This establishes interaction and audiovisual coupling as more than incidental presentation features (Levin, 2000a).

### 2.1 Artwork identity finding: 5, 6 or 7 systems?

Contemporary MIT documentation describes AVES as a set of **five** interactive systems (Levin, 2000a). NTT InterCommunication Center’s 2006 exhibition documentation describes a set of **six** interactive software systems (NTT ICC, 2006). Levin’s current archive describes a set of **seven** systems (Levin, n.d.). The discrepancy is not safely resolved by choosing the newest number.

**Finding 01 – Configuration ambiguity.** The 5–6–7 discrepancy may represent later additions, distinct iterations, exhibition-specific configurations, retrospective reclassification or documentary error. An authoritative collection record should preserve the discrepancy and investigate it through artist/studio consultation, original software inventories, exhibition records, acquisition documentation and version-specific technical evidence.

This issue demonstrates why artwork identity precedes technical treatment. A conservator cannot determine whether a component is missing until the intended configuration is known.

### 2.2 Audio development is itself historically significant

A contemporary Ars Electronica document states that a simple AVES installation comprised five Pentium computers, each with speakers and a mouse or preferably a Wacom pen tablet, while also noting that sonification of the works was still being completed in 2000 (Levin, 2000b). This complicates any assumption that every documented AVES configuration had identical audio behaviour at every point in its development. Module-level and version-level evidence is therefore required.

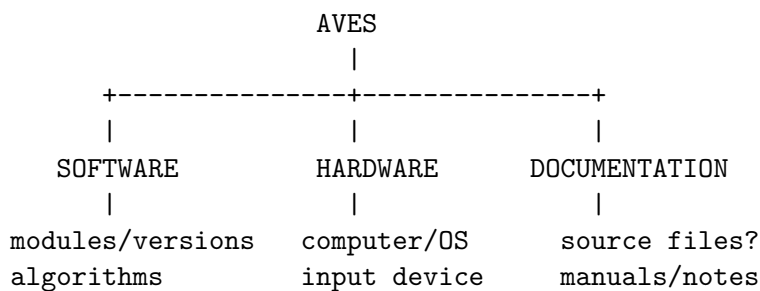
### 3 Significant Properties and Conservation Logic

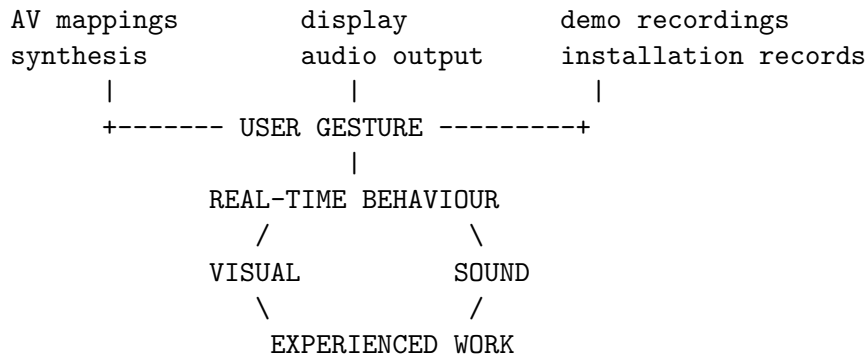
My initial prioritisation was **visual appearance, sound and interaction**. These are the principal experiential properties through which AVES is encountered. They are supported by lower-level technical properties such as software behaviour, input mapping, latency, rendering, synthesis behaviour, operating-system dependencies and hardware.

| Property                           | Priority               | Conservation rationale                                                                                                                                                                                                |
|------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visual appearance                  | Core                   | Abstract form, colour, motion and temporal behaviour constitute a primary output of the systems. A technically functional migration that substantially changes rendering may alter the work.                          |
| Sound                              | Core                   | AVES is conceived around tightly linked image and synthetic sound. For Yellowtail, screen-buffer brightness is mapped to amplitudes in an additive synthesis process (Levin, 2000c).                                  |
| Interaction                        | Core                   | The user is not simply selecting prerecorded media. Gestures create and manipulate the audiovisual output in real time. Recordings therefore preserve examples of performances, not the complete interactive artwork. |
| Software behaviour                 | Supporting / high      | Algorithms and mappings determine the relationship between gesture, image and sound.                                                                                                                                  |
| Latency / responsiveness           | Supporting / high      | Excessive latency could change the felt simultaneity between action and audiovisual response even if the final image and sound remain superficially similar.                                                          |
| Input hardware                     | Supporting / to verify | Historical evidence refers to mouse and preferably Wacom pen tablet. Whether a particular device is essential, preferred or replaceable requires artist/version-specific confirmation.                                |
| OS / rendering / audio environment | Supporting / high      | The historical software environment may condition timing, graphics and synthesis behaviour and is vulnerable to obsolescence.                                                                                         |

This hierarchy avoids treating every historical component as equally sacred. The conservation question is not simply whether a Pentium computer survives, but whether replacement or virtualised components can maintain the significant behaviours and experience of the work. Research on software-based art similarly emphasises significance, dependencies and virtualisation when mitigating technological obsolescence (Falcao, Dekker and Laurenson, 2016).

### 4 Component and Dependency Assessment





The map highlights a key distinction between *source components*, *enabling components* and *documentation*. A video can be essential evidence without being an executable constituent of the work. Conversely, a replaceable display device may not be conceptually unique but can still affect colour, scale, timing and perceived responsiveness.

## 5 Technical Examination of Historical Yellowtail Documentation

### 5.1 Provenance and status

A high-resolution QuickTime link labelled Yellowtail was retrieved from the historical MIT Media Laboratory AVES project page on 12 August 2026. The local file retained its retrieved filename, `yellowtail.mov`. It was treated as **historical documentation media**, not as an asserted artwork master or source file.

### 5.2 MediaInfo and FFprobe findings

MediaInfo and FFprobe were used independently to reduce reliance on a single parser. The following characteristics were observed:

| Characteristic                     | Observed value                                                                                 |
|------------------------------------|------------------------------------------------------------------------------------------------|
| Container                          | QuickTime / MOV                                                                                |
| Duration                           | 177.443443 seconds (approximately 2 min 57 s)                                                  |
| File size                          | 16,547,258 bytes                                                                               |
| Overall bit rate                   | 746,029 bit/s                                                                                  |
| Video codec                        | QuickTime Video (RPZA)                                                                         |
| Video dimensions                   | 240 x 180 pixels                                                                               |
| Display aspect ratio               | 4:3 (MediaInfo)                                                                                |
| Nominal / reported video rate      | approximately 29.97 fps                                                                        |
| Audio codec                        | PCM signed 16-bit big-endian ( <code>pcm_s16be</code> ; MediaInfo codec ID <code>twos</code> ) |
| Audio sampling rate                | 22,050 Hz                                                                                      |
| Audio channels                     | 1 (mono)                                                                                       |
| Audio bit depth                    | 16 bit                                                                                         |
| Audio bit rate                     | 352.8 kb/s                                                                                     |
| Encoded date reported by MediaInfo | 21 April 2000, 16:07:20 UTC                                                                    |

The audio figures are internally consistent: 22,050 samples/s x 16 bits x 1 channel = 352,800 bit/s. The documentation movie therefore contains a substantial uncompressed mono PCM stream. This does *not* establish that the artwork itself was intrinsically fixed to 22.05 kHz/16-bit mono output; it establishes the encoding characteristics of this retrieved documentation file.

### 5.3 Frame-rate anomaly and cross-validation

MediaInfo reported variable frame rate, a normal rate of 29.970 fps, and an anomalous maximum value of 1498.500 fps. FFprobe reported `r_frame_rate=2997/100` and an average rate also close to 30 fps. I therefore did not interpret 1498.5 fps as normal playback. It was retained as a timing/metadata anomaly that would warrant frame-level timestamp investigation if the documentation object itself were being accessioned or treated.

**Finding 02 – Cross-tool validation.** A surprising metadata value should not be silently discarded or accepted. The appropriate response is to record it, compare independent parsers, inspect timestamps if required, and distinguish parser/timing anomalies from observable playback behaviour.

### 5.4 Fixity and decode integrity

A SHA-256 baseline was established for the retrieved file:

```
cf6d868c58be832f315c94fa23280e7b06ae0a06fd51401a712c0837488b6530
```

A subsequent `shasum -a 256 -c` check returned OK. This supports bit-level identity relative to the recorded baseline; it does not prove audiovisual correctness, completeness or artwork authenticity. A full FFmpeg decode test at error-reporting level completed without reported decoding errors.

### 5.5 QCTools and perceptual review



Figure 1: QCTools overview generated during the independent examination of the retrieved `yellowtail.mov`. Screenshot: author, 12 August 2026.

QCTools displayed changing audio and video statistics across the file. Manual review showed that the principal graph changes corresponded to changes in the documentation content rather than

obvious faults. The movie combines explanatory material with demonstrations of the work: around 1:55, a more active audiovisual/rhythmic demonstration produces denser and more dynamic audio behaviour; around 2:25, a demonstration ends, the image transitions to black and narration returns, producing a comparatively steadier audio profile. Spot checks did not reveal obvious unintended freezes, audio dropouts or perceptible decode artefacts.

**Finding 03 – QC requires context.** Signal variation is not synonymous with damage. Quantitative graphs were used to locate moments for manual inspection, and the observed changes were interpreted in relation to programme structure and audiovisual content.

## 6 Risk Register

| Risk                                       | Impact   | Priority | Proposed response                                                                                                                                                      |
|--------------------------------------------|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artwork configuration ambiguity (5/6/7)    | High     | High     | Preserve contradictory evidence; establish version/iteration history; artist/studio interview; compare software inventories and exhibition records.                    |
| Loss of original binaries / source / media | Critical | Critical | Acquire and preserve original software carriers and binaries where rights permit; create forensic disk images; establish fixity; retain read-only preservation copies. |
| Windows 2000 dependency                    | High     | Critical | Document original environment; test virtualisation/emulation; retain original environment images; validate behaviour against reference evidence.                       |
| Legacy hardware / input dependency         | High     | High     | Inventory computers, graphics/audio hardware, mice/tablets and interfaces; distinguish dedicated from replaceable equipment; document calibration and mapping.         |
| Changed audiovisual timing or latency      | High     | High     | Establish reference behaviour through artist testimony, recordings and testing; measure and document response under emulation/migration.                               |
| Changed graphics rendering                 | High     | High     | Compare colour, motion, scaling and rendering between historical/reference and future environments.                                                                    |
| Changed audio synthesis/output             | High     | High     | Document synthesis logic, output path, channel configuration, levels and speaker requirements; compare future implementations perceptually and technically.            |
| Documentation loss                         | High     | High     | Consolidate artist statements, manuals, installation diagrams, exhibition records, photos, videos, version notes and conservation decisions.                           |
| Documentation media obsolescence           | Medium   | Medium   | Preserve original documentation file with fixity; create well-documented access/preservation derivatives when justified; retain provenance and transformation logs.    |



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|                     |      |      |                                                                                                                    |
|---------------------|------|------|--------------------------------------------------------------------------------------------------------------------|
| Rights restrictions | High | High | Establish rights to preserve, image, emulate, migrate, copy and exhibit software/media before treatment decisions. |
|---------------------|------|------|--------------------------------------------------------------------------------------------------------------------|

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## 7 Preservation Strategy

### 7.1 Why recordings are insufficient

A performance or demonstration recording captures a particular output. AVES, however, produces variable output through interaction. A recording cannot reproduce the user's capacity to create, manipulate and receive real-time audiovisual feedback. It is therefore reference/documentation material rather than a complete preservation surrogate.

### 7.2 Emulation as the first investigation, not an automatic solution

My first preservation option would be to investigate **emulation/virtualisation** of the historical Windows 2000 environment before rewriting the software. This is a conservative starting point because it seeks to retain original executables and their historical logic. Tate's software-art practice identifies disk imaging and emulation among standardised preservation workflows for software-based works (Ensom and Falcao, 2022).

However, a successful boot is not equivalent to a successful conservation outcome. An emulated AVES environment would need validation against significant properties: gesture mapping, responsiveness, latency, visual rendering, sound synthesis and output, and any version-specific hardware behaviour. If these cannot be maintained, migration or partial reimplementations may become necessary, but such interventions should be documented and justified against the work's identity rather than treated as routine upgrades.

### 7.3 Recommended preservation package

For a hypothetical acquisition, I would seek to assemble:

- original software binaries and, where available and legally transferable, source code;
- forensic disk/carrier images and checksums;
- operating-system and dependency information;
- module/version inventory tied to the 5/6/7 identity question;
- hardware and input-device inventory, including replaceability status;
- display and audio specifications, wiring and calibration information;
- installation and exhibition records;
- artist/studio interview addressing change tolerances and essential properties;
- reference audiovisual documentation and screenshots;
- emulation/migration test records and comparison evidence;
- rights documentation covering preservation copying, emulation/migration and exhibition.

This approach is consistent with Matters in Media Art's emphasis on component verification, preservation material, display specifications, artist information, structure/condition assessments and ongoing migration records (Matters in Media Art, n.d.).

## 8 Questions for an Artist / Studio Interview

The 5–6–7 discrepancy makes artist consultation particularly important. Priority questions would include:

1. Which systems constitute AVES in its 1998–2000 form, and when were additional systems incorporated or reclassified?
2. Does "AVES" identify a fixed suite, an evolving family of instruments, or configurations selected for particular exhibitions?

3. Which modules and versions are essential for an installation to be described as AVES?
4. Which historical input devices, displays, computers and audio systems are essential, preferred or replaceable?
5. What degree of change in latency, frame timing, colour, resolution or sound is acceptable?
6. Is emulation acceptable? Under what circumstances would migration or code modification be acceptable?
7. Are there reference configurations, source files, installers, disk images, manuals or undocumented dependencies that should accompany preservation?
8. How should future conservators judge whether audiovisual mapping and responsiveness remain authentic?

## 9 Reflection on the Assessment Process

The practical exercise changed the way I understood the task. At first, the problem appeared close to recording and safely storing an electronic artwork. The investigation showed that the more difficult question is determining what must remain reproducible for the artwork to retain its identity. For AVES, the output file is not the work: the relationship between gesture, software behaviour, image and sound is fundamental.

The technical tools were useful because they produced evidence rather than conclusions. MediaInfo quickly exposed technical metadata; FFprobe independently clarified stream and codec structure; SHA-256 established a fixity baseline; FFmpeg tested decode behaviour; and QCTools located signal changes for perceptual review. The most important step was interpreting these outputs in context. For example, the QCTools graph changed substantially during the movie, but manual review showed that these changes aligned with transitions between narration and audiovisual demonstration. Similarly, an extreme maximum-frame-rate value was not accepted at face value but retained as an anomaly after cross-validation.

The artwork-identity research was equally important. The 5–6–7 discrepancy demonstrates that conservation documentation must preserve uncertainty rather than create false precision. Before deciding what to emulate, migrate or replace, a museum would need to establish which iteration it owns and which behaviours and components define that iteration.

## 10 Conclusion

AVES is a useful case study because it demonstrates the convergence of software preservation, audiovisual engineering, interaction design and museum documentation. The work's long-term viability depends not only on preserving bits but on maintaining a defensible relationship between historical software, user gesture, visual behaviour and synthetic sound.

The strongest immediate priorities are therefore to resolve the suite's version/configuration history, secure original software and technical documentation, establish fixity and provenance, document hardware and audiovisual dependencies, and test an emulation-first strategy against clearly defined significant properties. Historical recordings such as the Yellowtail movie remain valuable reference evidence, but they cannot substitute for the interactive generative system.

## References

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## A Technical Evidence Log

| Action           | Evidence / result                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source retrieval | <code>yellowtail.mov</code> , retrieved from historical MIT AVES page, 12 August 2026; treated as documentation media.                                                                                                                            |
| MediaInfo        | Human-readable TXT and XML metadata outputs created.                                                                                                                                                                                              |
| FFprobe          | JSON stream/container record created; RPZA video and PCM signed 16-bit big-endian mono audio identified.                                                                                                                                          |
| Fixity           | SHA-256: <code>cf6d868c58be832f315c94fa23280e7b06ae0a06fd51401a712c0837488b0</code> subsequent check returned OK.                                                                                                                                 |
| Decode test      | Full FFmpeg decode completed with no error-level messages recorded.                                                                                                                                                                               |
| QCTools          | File parsed; overview reviewed; perceptual spot checks at approximately 00:00:50, 00:01:55 and 00:02:25. Changes interpreted in relation to narration/demonstration structure; no obvious unintended freeze/dropout identified at sampled points. |

## B Suggested Project Folder Structure

```

AVES_TBM_CaseStudy/
  01_source_documentation/
    yellowtail.mov
  02_metadata/
    yellowtail_medaiinfo.txt
    yellowtail_medaiinfo.xml
    yellowtail_ffprobe.json
  03_fixity/
    yellowtail_sha256.txt
  04_qc/
    yellowtail_decode_errors.txt
  05_analysis/

```

```
artwork_identity_notes.txt  
06_portfolio_assets/
```

## C Limitations

This study did not examine original AVES executables, source code, disk images, installation masters, hardware, an emulated Windows 2000 instance or an artist interview. No claim is made that the technical characteristics of `yellowtail.mov` are the technical specifications of the live artwork. Risk ratings and preservation recommendations are therefore preliminary and would need revision if collection-level materials became available.